

Experiment No.11

Aim: To study and implement container orchestration using Kubernetes on AWS/Azure/Google cloud platform.

Objectives:

- Develop a theoretical understanding of container orchestration principles, focusing on Kubernetes as a powerful orchestration tool for managing containerized applications. Apply practical knowledge to configure and utilize Kubernetes on AWS, emphasizing cluster creation, deployment, scaling, and management.

Outcomes:

- To demonstrate proficiency in utilizing Kubernetes for container orchestration on the AWS cloud platform.
- Apply knowledge in creating, scaling, and managing Kubernetes clusters on AWS, understanding the benefits of container orchestration for complex application deployments.

Software Requirements: Windows/Ubuntu OS, AWS Account, Docker, Kubernetes.

Theory:

Kubernetes is an open-source container orchestration platform that automates the deployment, scaling, and management of containerized applications. It provides features such as service discovery, load balancing, rolling updates, and self-healing capabilities. Amazon Elastic Kubernetes Service (EKS) is an AWS-managed Kubernetes service that simplifies the deployment and management of Kubernetes clusters.

Kubernetes architecture includes nodes (worker machines) and a control plane (master nodes). Nodes host the containers, and the control plane manages the overall cluster. The kubectl command-line tool is used to interact with the Kubernetes cluster and deploy applications.

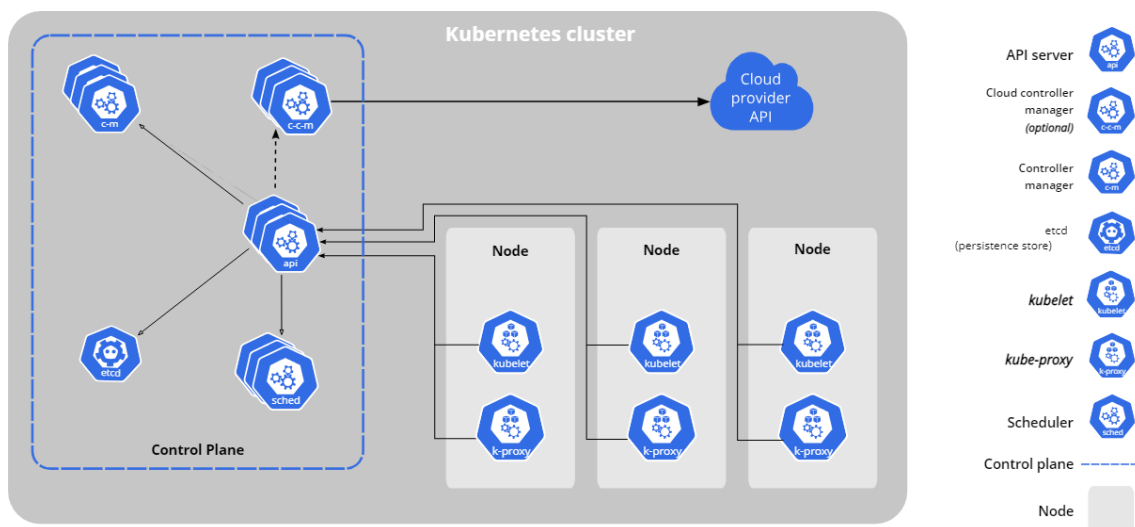


Figure 11 Kubernetes architecture

Key Components:

- **Master Nodes:** The control plane responsible for managing the cluster, including:
 1. **API Server:** Interface for interacting with Kubernetes, receiving commands and returning cluster state information.
 2. **Scheduler:** Assigns containers to nodes based on resource requirements and availability.
 3. **Controller Manager:** Ensures the cluster state matches the desired state, handling events like container failures and node additions.
 4. **Etcd:** Distributed key-value store for storing cluster configuration data.
- **Worker Nodes:** Hosts that run containerized applications, consisting of:
 1. **Kubelet:** Agent responsible for managing containers on the node, communicating with the API server.
 2. **Kube-proxy:** Network proxy for managing network rules and forwarding traffic to containers.
- **Pods:** The smallest deployable unit in Kubernetes, encapsulating one or more containers and shared resources.
- **Services:** Abstract way to expose a set of pods as a network service, enabling load balancing and discovery.

How Kubernetes Works:

1. **Deployment:** You define desired cluster state using YAML manifests (e.g., pods, services).
2. **Scheduling:** The scheduler places pods on available worker nodes based on resource needs and constraints.
3. **Container Management:** Kubelet starts and manages containers on nodes, ensuring they run as specified.
4. **Health Checks:** Kubernetes monitors container health and automatically restarts or replaces failed containers.
5. **Networking:** Kube-proxy configures networking rules for pods and services, enabling communication between them.
6. **Scaling:** Kubernetes can automatically scale the number of pods based on resource usage or other metrics.
7. **Self-Healing:** Kubernetes automatically recovers from node failures or other disruptions.

Benefits of Elastic Beanstalk:

- **Automation:** Automates deployment, scaling, and management of containerized applications.
- **High Availability:** Ensures applications remain available even in the event of node failures.
- **Resource Efficiency:** Optimizes resource utilization across the cluster.
- **Scalability:** Easily scales applications horizontally by adding more nodes.

Stepwise Procedure to Implement PaaS using AWS Elastic Beanstalk:

Step 1: Sign into AWS Console

- Log in to the AWS Management Console using your AWS account credentials.

Step 2: Create EKS Cluster

- Navigate to the Amazon EKS service in the AWS Console.
- Create an EKS cluster, specifying cluster details and networking configurations.

Step 3: Configure kubectl

- Install kubectl on the local machine.
- Configure kubectl to connect to the EKS cluster.

Step 4: Deploy Sample Application

- Create Kubernetes deployment files for a sample application.
- Deploy the sample application to the EKS cluster using kubectl apply.

Step 5: Scale and Manage Deployments

- Scale the number of running pods in the deployment.
- Update the application and observe rolling updates.

Step 6: Deploy Services and Ingress

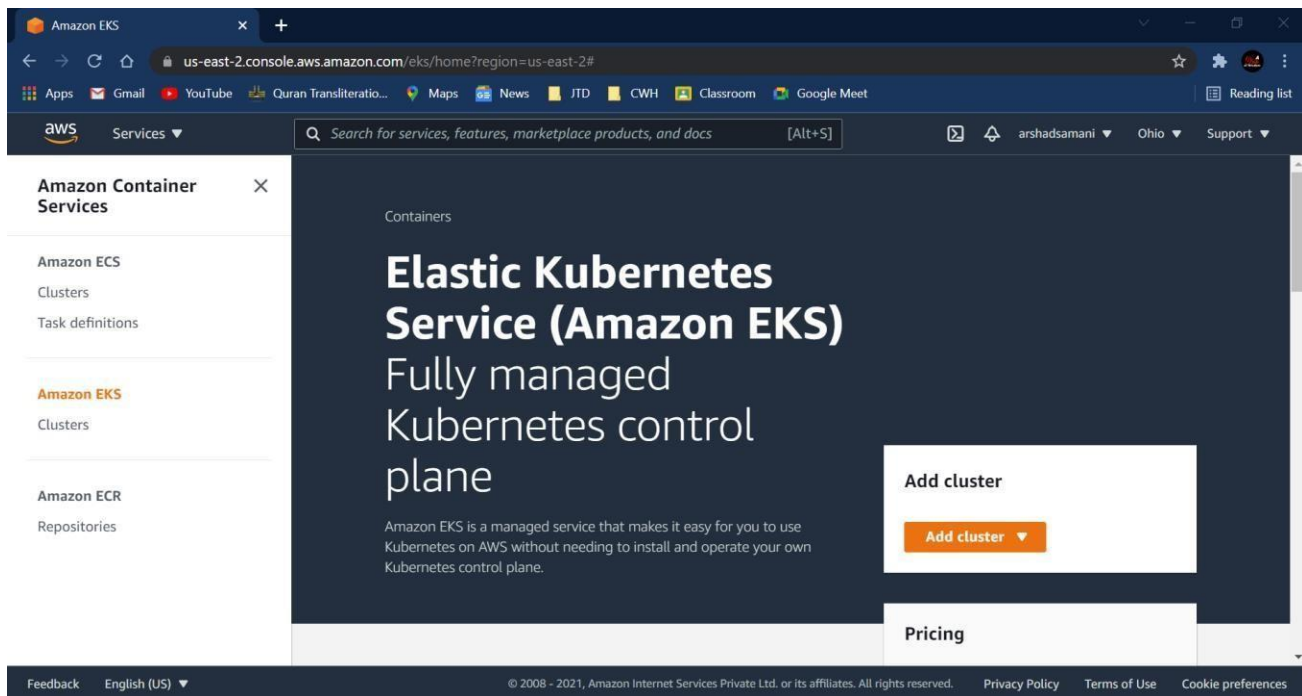
- Create Kubernetes service and ingress files.
- Deploy services and ingress resources to expose the application externally.

Step 7: Monitor and Troubleshoot

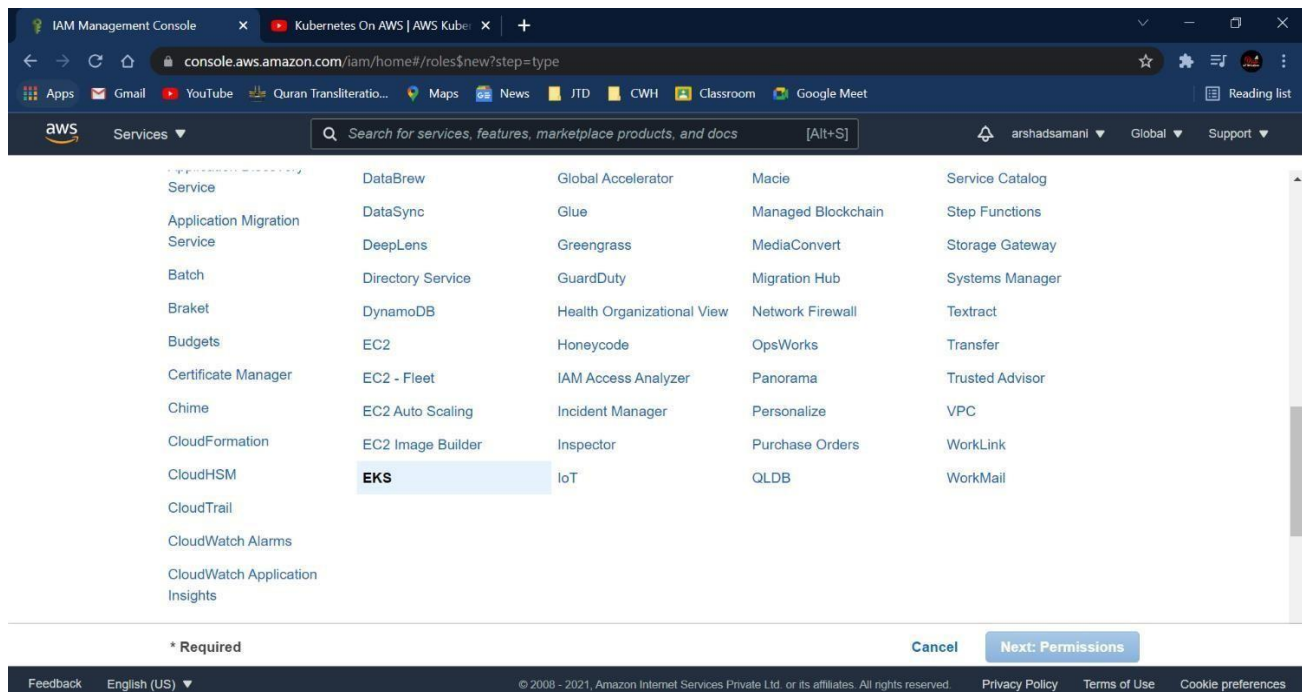
- Use kubectl to monitor the status of pods, deployments, and services.
- Troubleshoot common issues and observe Kubernetes logs.

Result:

1. Starting with Kubernetes.



2. Selecting EKS



3. Select EKS - Cluster

The screenshot shows the AWS IAM Management Console with the 'Kubernetes On AWS | AWS Kube' tab active. The URL is `console.aws.amazon.com/iam/home#/roles$new?step=type`. The 'Select your use case' section lists several options, with 'EKS - Cluster' selected and highlighted in blue. Below the list, there are buttons for 'Cancel' and 'Next: Permissions'. The footer includes 'Feedback', 'English (US)', and copyright information.

Insights

Select your use case

EKS
Allows EKS to manage clusters on your behalf.

EKS - Cluster
Allows access to other AWS service resources that are required to operate clusters managed by EKS.

EKS - Connector
Allows access to other AWS service resources that are required to connect to external clusters.

EKS - Fargate pod
Allows access to other AWS service resources that are required to run Amazon EKS pods on AWS Fargate.

EKS - Fargate profile
Allows EKS to run Fargate tasks.

EKS - Nodegroup
Allow EKS to manage nodegroups on your behalf.

* Required

Cancel Next: Permissions

Feedback English (US) © 2008 - 2021, Amazon Internet Services Private Ltd. or its affiliates. All rights reserved. Privacy Policy Terms of Use Cookie preferences

4. Creating Role

The screenshot shows the 'Create role' page in the AWS IAM Management Console, specifically the 'Review' step. The URL is `console.aws.amazon.com/iam/home#/roles$new?step=review&selectedService=AmazonEKS&selectedUseCase=AmazonEKSClusterPolicy`. The 'Role name' is 'eksrole' and the 'Role description' is 'Allows access to other AWS service resources that are required to operate clusters managed by EKS.' The 'Trusted entities' are 'AWS service: eks.amazonaws.com' and the 'Policies' are 'AmazonEKSClusterPolicy'. There are buttons for 'Cancel', 'Previous', and 'Create role'. The footer includes 'Feedback', 'English (US)', and copyright information.

Create role

Review

Provide the required information below and review this role before you create it.

Role name* eksrole
Use alphanumeric and '+', '@', '-' characters. Maximum 64 characters.

Role description Allows access to other AWS service resources that are required to operate clusters managed by EKS.
Maximum 1000 characters. Use alphanumeric and '+', '@', '-' characters.

Trusted entities AWS service: eks.amazonaws.com

Policies AmazonEKSClusterPolicy

* Required

Cancel Previous Create role

Feedback English (US) © 2008 - 2021, Amazon Internet Services Private Ltd. or its affiliates. All rights reserved. Privacy Policy Terms of Use Cookie preferences

5. Role has been Created

The screenshot shows the AWS IAM Management Console. A green notification banner at the top states "The role eksrole has been created." The left sidebar shows the "Roles" link under "Access management". The main content area displays a list of roles, including "AWSServiceRoleForAWSCloud9", "AWSServiceRoleForOrganizations", and "AWSServiceRoleForSupport".

Role name	Trusted entities	Last acti...
AWSServiceRoleForAWSCloud9	AWS Service: cloud9 (Service-Linked Role)	21 days ago
AWSServiceRoleForOrganizations	AWS Service: organizations (Service-Linked Role)	-
AWSServiceRoleForSupport	AWS Service: support (Service-Linked Role)	-

6. Cloud Formation

The screenshot shows the AWS CloudFormation console. The left sidebar shows the "Stacks" link under "CloudFormation". The main content area displays a list of stacks, including "aws-cloud9-npm-user-b391cb4d24e648e98f9bac22dbf8ca6" with a status of "CREATE_COMPLETE".

Stack name	Status	Created time	Description
aws-cloud9-npm-user-b391cb4d24e648e98f9bac22dbf8ca6	CREATE_COMPLETE	2021-10-07 14:50:23 UTC+0530	-

CloudFormation - Stack

Kubernetes On AWS | AWS Kube

Getting started with Amazon EK

us-east-2.console.aws.amazon.com/cloudformation/home?region=us-east-2#/stacks/create/template

Apps Gmail YouTube Quran Transliteration... Maps News JTD CWH Classroom Google Meet

aws Services

Search for services, features, marketplace products, and docs [Alt+S]

arshadsamani Ohio Support

Configure stack options

Step 4 Review

in the stack.

Template is ready

Use a sample template

Create template in Designer

Specify template

A template is a JSON or YAML file that describes your stack's resources and properties.

Template source

Selecting a template generates an Amazon S3 URL where it will be stored.

Amazon S3 URL

Upload a template file

Amazon S3 URL

https://amazon-eks.s3.us-west-2.amazonaws.com/cloudformation/2020-10-29/amazon-eks-vpc-private-subnets.yaml

Amazon S3 template URL

S3 URL: https://amazon-eks.s3.us-west-2.amazonaws.com/cloudformation/2020-10-29/amazon-eks-vpc-private-subnets.yaml

View in Designer

Cancel

Next

Feedback

English (US)

© 2008 - 2021, Amazon Internet Services Private Ltd. or its affiliates. All rights reserved. Privacy Policy Terms of Use Cookie preferences

CloudFormation - Stack

Kubernetes On AWS | AWS Kube

Getting started with Amazon EK

us-east-2.console.aws.amazon.com/cloudformation/home?region=us-east-2#/stacks/create/parameters

Apps Gmail YouTube Quran Transliteration... Maps News JTD CWH Classroom Google Meet

aws Services

Search for services, features, marketplace products, and docs [Alt+S]

arshadsamani Ohio Support

CloudFormation > Stacks > Create stack

Step 1 Specify template

Step 2 Specify stack details

Step 3 Configure stack options

Step 4 Review

Specify stack details

Stack name

Stack name

eks-service

Stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).

Parameters

Parameters are defined in your template and allow you to input custom values when you create or update a stack.

PublicSubnet01Block

CidrBlock for public subnet 01 within the VPC

192.168.0.0/18

PublicSubnet02Block

CidrBlock for public subnet 02 within the VPC

192.168.64.0/18

PrivateSubnet01Block

CidrBlock for private subnet 01 within the VPC

192.168.128.0/18

PrivateSubnet02Block

CidrBlock for private subnet 02 within the VPC

192.168.192.0/18

Cancel

Previous

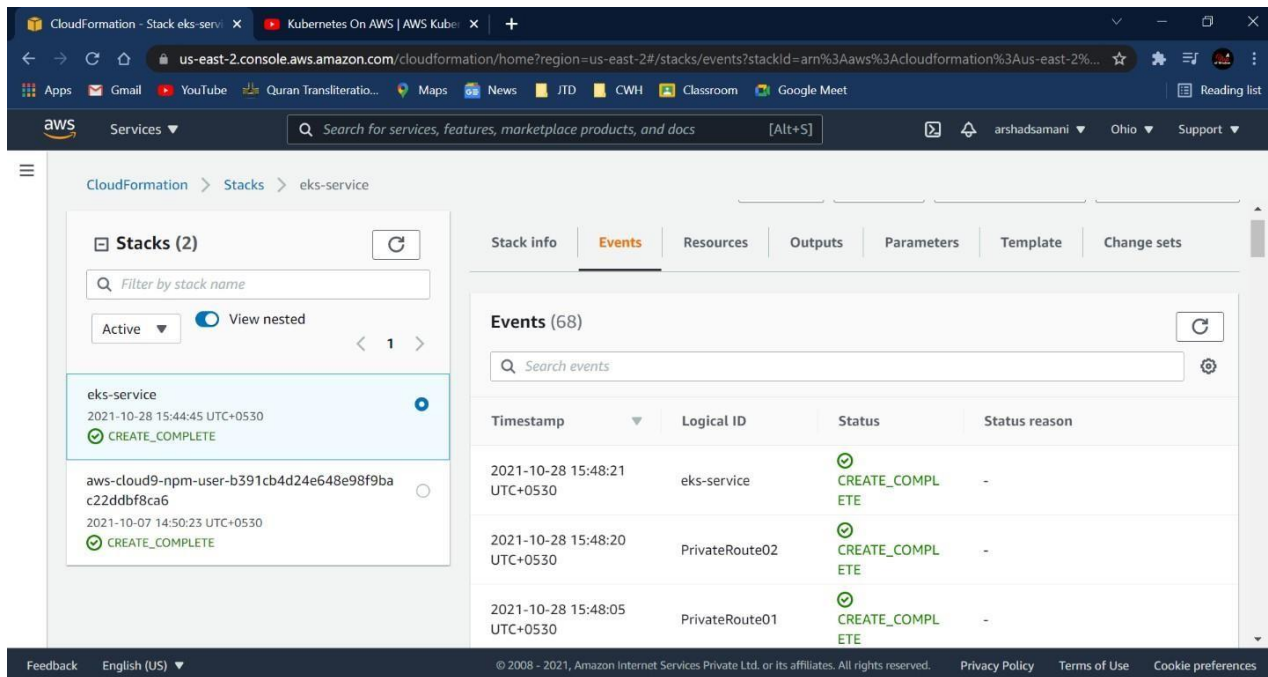
Next

Feedback

English (US)

© 2008 - 2021, Amazon Internet Services Private Ltd. or its affiliates. All rights reserved. Privacy Policy Terms of Use Cookie preferences

7. Stack is Created



CloudFormation > Stacks > eks-service

Stacks (2)

Filter by stack name

Active View nested

eks-service
2021-10-28 15:44:45 UTC+0530
CREATE_COMPLETE

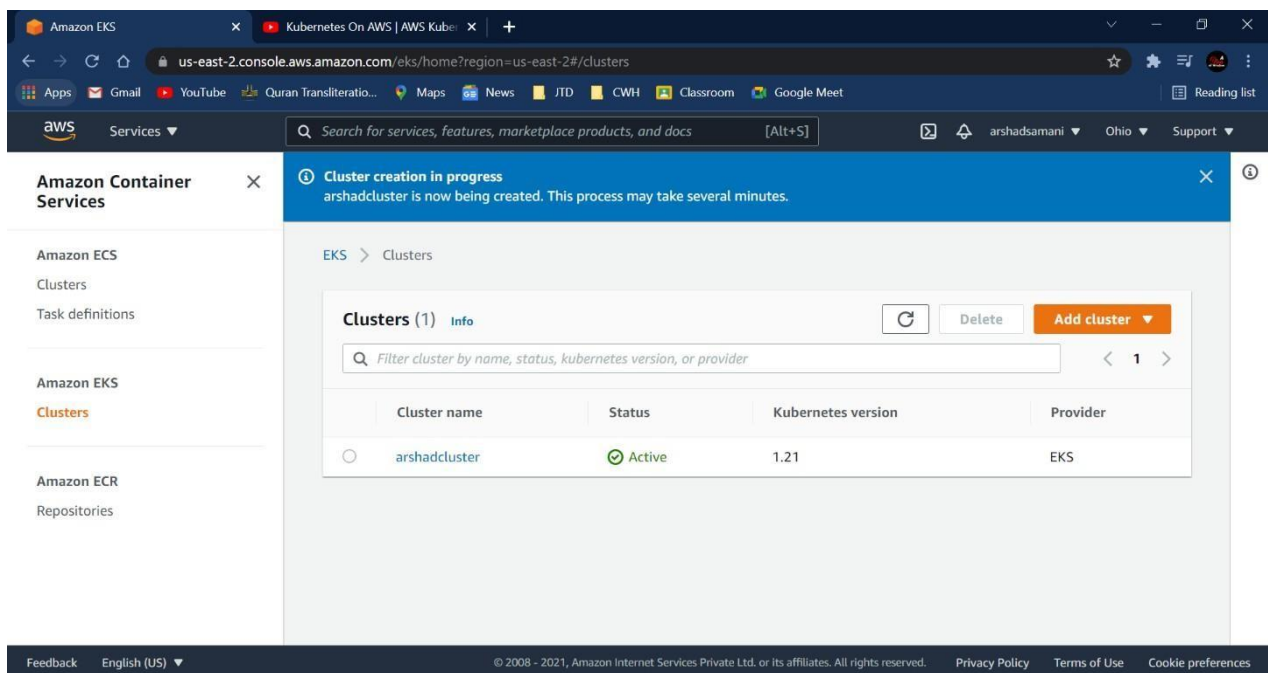
aws-cloud9-npm-user-b391cb4d24e648e98f9ba
c22ddb8ca6
2021-10-07 14:50:23 UTC+0530
CREATE_COMPLETE

Events (68)

Search events

Timestamp	Logical ID	Status	Status reason
2021-10-28 15:48:21 UTC+0530	eks-service	CREATE_COMPLETE	-
2021-10-28 15:48:20 UTC+0530	PrivateRoute02	CREATE_COMPLETE	-
2021-10-28 15:48:05 UTC+0530	PrivateRoute01	CREATE_COMPLETE	-

8. Cluster Created



Amazon EKS

us-east-2.console.aws.amazon.com/eks/home?region=us-east-2#/clusters

Cluster creation in progress
arshadcluster is now being created. This process may take several minutes.

EKS > Clusters

Clusters (1)

Filter cluster by name, status, kubernetes version, or provider

Cluster name	Status	Kubernetes version	Provider
arshadcluster	Active	1.21	EKS

9. Installing kubectl

Administrator: Windows PowerShell

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Writing web request
Writing request stream... (Number of bytes written: 16736610)
```

Administrator: Windows PowerShell

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell

PS C:\Windows\system32> curl -o kubectl.exe.sha256 https://amazon-eks.s3.us-west-2.amazonaws.com/1.21.2/2021-07-05/bin/windows/amd64/kubectl.exe.sha256
PS C:\Windows\system32> curl -o kubectl.exe.sha256 https://amazon-eks.s3.us-west-2.amazonaws.com/1.21.2/2021-07-05/bin/windows/amd64/kubectl.exe.sha256
PS C:\Windows\system32> Get-FileHash kubectl.exe

Algorithm      Hash
-----
SHA256         B5EC9989B31219241F205D1E2861F35A4C4515BC70DCC307225A0736EAF80C6
Path
C:\Windows\system32\kubectl.exe

PS C:\Windows\system32>
```

10. Successfully Installed

Administrator: Windows PowerShell

```
PS C:\Windows\system32> kubectl version --short --client
Client Version: v1.21.2-13+d2965f0db10712
PS C:\Windows\system32>
```

Conclusion:

Thus, we have successfully understood and implement container orchestration using Kubernetes on the AWS platform..